

CCT Campaign Plan — Fix $\hat{A}_i$, Find the Squid vs Jackal

Prepared by COMPASS (for Charles, SCOUT, Alex)
Date: 2026-08-21
Status: Act II sensitivity sweep complete (2026-08-21) — P1 full NO, P1 +DFT PPM YES (~2 pp, overlapping CIs); P3 shipped; next: Priority 2 heatmap (Charles/COMPASS)
SCOUT review: Complete (2026-08-21) — 20260821_SCOUT_CCT_campaign_review.md
Canonical companions (do not re-derive from memory):

Doc	Role
20260820_COMPASS_Charles_CCT_porch_reading.md	Narrative + science (SCOUT approved 2026-08-21)
20260821_SCOUT_porch_reading_review.md	Spot-checks, three clarifications
../HEROs_and_PASSEs/PD20_22_campaign_big_picture.md	Dissertation arc (Army → MBB ladder)
../../BINDING_Selection_is_its_own_step.md	Score $\neq$ select $\neq$ environment

0 — One sentence (what Alex just got excited about)

Hold individual talent $\hat{A}_i$ fixed. Then ask: at the same fish size, does draft probability fall when the player moves from a Squid pond (stands out on a pretty-good team) to a Jackal pond (same talent, crowded among contenders)?
That is the Central Contention and Theme (CCT). Alex's reaction — fix A_i — is exactly the microscope this campaign builds. Marginal hero curves average over ability; they cannot answer this question. Conditional plots can.

1 — Campaign arc (three acts)

flowchart LR; A[Act I – BDP population geometry] --> B[Act II – Conditional CCT empirics]; B --> C[Act III – HAND deck + Alex defense]; A -. -> IDONEI A; B --> D[Act IIb – Sensitivity optional]; D --> C;

Act	Question	Status	Owner
I — BDP	What do the pools look like? ($\hat{A}$, T_j , roster size, +DFT)	Closed (2026-08-20)	Charles + scripts
II — Conditional	At fixed $\hat{A}$, does draft rate fall as pond thickens?	Sweep done — see §3 scorecard; whisper in +DFT PPM only	SCOUT · Charles (HAND)
III — HAND + Alex	Can you show and defend CCT in one deck?	After Act II figures	Charles (slides) · COMPASS (wording)

Win condition (revised 2026-08-21): A defensible HAND story — not a single pooled bar — that Charles can read to Alex without BINDING violations. Partial win: PPM +DFT shows Squid > Jackal at matched $\hat{A}$ (~23% vs ~21%, overlapping CIs). Full-panel P1 and OBPM do not. Priority 3 adds within-team rank on contenders. Priority 2 heatmap may localize where the +DFT whisper lives.
Not the win condition: Bin 16 dips on a single ventile curve; claiming CCT confirmed from +DFT alone without caveats; promoting OBPM to canonical.

2 — Scientific guardrails (binding)

- Environment ($L_{\text{net}} = B - D$) $\neq$ advancement. CCT is a draft / selection story, not “bad teammates hurt my box score.”
 - Score ($S_i = A_i - \lambda L_C$) $\neq$ select (top-K, Gibbs). Hero is an outcome; do not merge layers in talk or captions.
 - Congestion axis: poolq_loo first (“teammates around me,” excluding self). T_j second only — team mean includes the player.
 - Locked panel for claims: mg10 min20 11_21 (same estimand as POST-QC hero). QC always on: drop_dash_placeholder_names=True (dash “-” rows). FP / min0 = QC story only.
 - Perf metric: PPM canonical; BPM / OBPM = robustness appendix (Track C: no elite dip rescue).
 - +DFT overlay: magnifier on draft-ecosystem subsample — not “drafted players only” on player histograms.
- Army cross-read (do not garble): Army screams (robust macro tail drop). Captain promotion ~35–40% — assembly-line baseline, not tiny K/N. NCAA draft ~2–2.5%. Army loud $\neq$ Army slots scarcer than NBA picks.

3 — Plot menu (priority order)

Build in order unless Charles reorders. Each plot ships PNG + JSON sidecar (cell counts mandatory).

Priority 1 — Matched $\hat{A} \times$ pond (THE Alex plot) ★

Spec

- Panel: mg10 min20 11_21

- Fix narrow $\hat{A}_i$ band (default: $z \in [1.5, 2.0]$; sensitivity: top decile)
- X-axis: poolq_loo ventiles (preferred) or T_j ventiles
- Y-axis: draft rate (with binomial CI if n allows)
- poolq winsor: 0.01–0.99 (match locked POST-QC hero — BDP $\hat{A}/T$ plots omit winsor; this plot does not)
- Squid proxy: mid pond / mid T_j · Jackal proxy: top pond / top T_j — same $\hat{A}$ band

CCT signature: Jackal bar below Squid bar at same individual talent.

Outputs

- basic_data_plots/CCT_draft_rate_ai_band_poolq_loo.png
- basic_data_plots/CCT_draft_rate_ai_band_poolq_loo.json

Priority 1 scorecard (2026-08-21) — all variants shipped

Spec locked: mg10 min20 2011–2021 · PPM/OBPM $z \in [1.5, 2.0]$ · poolq_loo ventiles (within band) · winsor 0.01–0.99 · Squid = bins 6–8 · Jackal = bins 14–16.

Variant	Squid	Jackal	CCT (Squid > Jackal)?	Files
PPM · full panel	7.3% (n=382)	10.8% (n=381)	NO	CCT_draft_rate_ai_band_poolq_loo.*
PPM · +DFT	22.9% (n=140)	20.9% (n=139)	YES (~2 pp; CIs overlap)	...poolq_loo_dft.*
OBPM · full	16.3% (n=306)	37.6% (n=306)	NO (strong opposite)	...poolq_loo_obpm.*
OBPM · +DFT	29.0% (n=114)	39.8% (n=113)	NO	...poolq_loo_obpm_dft.*
PPM · T_j axis (8 bins)	—	—	Draft rate rises with T_j at fixed $\hat{A}$	CCT_draft_rate_fixedAi_Tj*_Tj.*

Band (PPM full): 2,035 PS / 178 drafted (~8.8% in band; 166/178 drafts on +DFT teams). All 16 ventiles $n \approx 127$ –128 (full) or ~46 (+DFT).

[COMPASS] synthesis

1. Full league: thicker poolq_loo → higher draft rate at fixed $\hat{A}$ — opposite naive CCT on this axis. Confound: elite programs raise both pond quality and draft visibility.
2. +DFT PPM: sign flips toward CCT — BDP “draft-ecosystem zoom” intuition validated, but effect is small and noisy (bin 14 spikes; bins 15–16 fall; pooled proxies hide ventile mess).
3. OBPM: wrong magnifier — Jackal dominates even more; appendix only (Track C).
4. ** T_j secondary:** monotonic rise with team mean — “better team helps” at fixed z .
5. Priority 3 (below): different question — within-team rank on Q4 contenders (top 9.5% vs mid 3.0%).

Honest Alex line (post-sweep): “Full panel at fixed talent: crowded ponds draft more. Restrict to programs that actually produce picks, PPM sign flips — mid-pond edges top-pond by ~2 points but uncertainty overlaps. OBPM goes the other way. On contenders, you must be near the top of your roster.”

HAND lead pair: side-by-side PPM full vs PPM +DFT + Priority 3 Q4 panel.

Priority 2 — $\hat{A} \times$ poolq_loo heatmap — NEXT

Draft rate in every cell ($\hat{A}$ ventile $\times$ poolq_loo ventile). Hero curve = column average — shows smearing. Look for upper-right (high A, high pond) cooler than upper-middle. Run on full panel and +DFT if feasible.

Wider bins at top if sparse (~1,100 total drafts). Flag cells with $n < 30$ in JSON.

Priority 3 — Within-team rank $\times T_j$ quartile — SHIPPED

Draft rate vs roster percentile, faceted by T_j quartile. Files: CCT_draft_rate_roster_pct_by_tj_quartile.{png,json}.

Result (2026-08-21): Q4 (highest T_j) — top roster rank 9.5% vs mid rank 3.0% (top > mid: YES). All quartiles show top rank > mid rank. Does not fix $\hat{A}$ — complements P1; supports “stand out on your team,” especially on contenders.

Priority 4 — Interval overlap + draft (HAND17 extension)

Color drafted vs not on interval slide logic; top T_j quartile; drafted players on leading edge of team interval?

Priority 5 — Model-aligned diagnostics

- L_j^C vs T_j by ventile
- MLE residuals: observed – predicted draft rate by poolq_loo at fixed A

Act IIb — Sensitivity — MOSTLY DONE

Variant	Status	Result
+DFT subsample (PPM P1)	✓	CCT YES, fragile (~2 pp)
OBPM full + +DFT	✓	CCT NO both; appendix only
T_j secondary panel	✓	Rate rises with team mean
mg10 min10	—	Not run
Alternate $\hat{A}$ bands	—	Charles call after P2 heatmap

4 — Engineering plan

New script (SCOUT)

Name (locked): `sports/scripts/pass_a_congestion_conditional.py`

Reuse: `pass_a_empirical_bundle.py` panel path, `pd20_22_campaign_window`, `hero_gallery_paths`, BDP layout helpers from `bdp_ai_tj_distributions.py` where sensible.

CLI sketch

```
python sports/scripts/pass_a_congestion_conditional.py \
  --season-min 2011 --season-max 2021 \
  --min-minutes 20 \
  --min-team-season-games 10 \
  --winsor-lo 0.01 --winsor-hi 0.99 \
  --ai-lo 1.5 --ai-hi 2.0 \
  --plot matched_pond \
  --out-dir 3-Master_Plan/re_entry/HEROs_and_PASSes/basic_data_plots
```

[SCOUT] `mg10` = `min_team_season_games 10` in code (drop team-seasons with ≤ 10 games; keep 11+). Do not pass `11` — that is not the repo knob name.

Incremental writes: JSONL or per-plot JSON with `n`, `drafts`, `rate`, `CI` per cell — append-friendly if we add bands later (`../../../../cursor/rules/incremental-writes.mdc`).

Do not clobber: Locked Pass A hero PNGs in `pass_a/`.

HAND deck (Charles)

New deck or new section in next HAND — working title `CHAR_CCT_HAND` or section in PD23 deck. **Change Picture** from `basic_data_plots/` and `slides/Basics_data_plots_HAND/` exports. Agents **never** overwrite `.pptx`.

5 — Checklist (Charles checkboxes only)

How to use: Charles changes `[ ]` → `[x]` and fills **Proof**. SCOUT/COMPASS update **Status** rows in §6.

Phase 0 — Green light

Done	Item	Proof
[]	Charles read porch doc + this campaign plan	Date
[x]	SCOUT reviewed this campaign plan	<code>20260821_SCOUT_CCT_campaign_review.md</code>
[x]	Priority 1 spec locked: $\hat{A}$ band, axis (<code>poolq_loo</code> vs <code>T_j</code>), season window	<code>poolq_loo</code> · <code>z</code> ∈ [1.5, 2.0] · 2011–2021 · <code>mg10 min20</code> · winsor 0.01–0.99
[x]	Charles says go on script build	2026-08-21 — Alex briefed; green-light Priority 1

Phase 1 — Priority 1 figure (Act II core)

Done	Item	Proof
[x]	<code>pass_a_congestion_conditional.py</code> exists, smoke test runs	<code>python sports/scripts/pass_a_congestion_conditional.py --plot matched_pond ...</code> exit 0 (2026-08-21)
[x]	Matched $\hat{A}$ × <code>poolq_loo</code> PNG + JSON on disk	<code>basic_data_plots/CCT_draft_rate_ai_band_poolq_loo.{png,json}</code>
[x]	JSON cell counts reviewed — no empty claims from <code>n < 10</code> cells	All 16 bins $n \approx 127\text{--}128$; band $n=2,035$ / 178 drafts; CCT=NO (Jackal 10.8% > Squid 7.3%)
[]	Charles eyeballs figures — Squid/Jackal test + sweep story	See §3 scorecard
[x]	COMPASS slide caption draft (score ≠ select language)	§7 (two-panel + P3)

Phase 2 — Act II expansion

Done	Item	Proof
[]	Priority 2 heatmap (full + +DFT if feasible)	NEXT — green-light SCOUT
[x]	Priority 3 roster percentile faceted	<code>CCT_draft_rate_roster_pct_by_tj_quartile.{png,json}</code>
[x]	+DFT subsample rerun of Priority 1	<code>CCT_draft_rate_ai_band_poolq_loo_dft.*</code>
[x]	OBPM robustness (full + +DFT)	<code>___obpm.*</code> , <code>___obpm_dft.*</code> — appendix only

Phase 3 — Act III (Alex-ready)

Done	Item	Proof
[]	HAND deck section assembled	<code>.pptx</code> path
[]	Alex paragraph rehearsed (§8 below)	PD # or date
[]	BINDING rule visible on one slide or speaker note	Slide #
[]	Campaign doc Status updated to Closed or Act III complete	This file

6 — Ownership & delegation

Task	Owner	Delegate?
Campaign sequencing, checklist, Alex wording	COMPASS	—
Script build, plots, JSON spot-checks	SCOUT	—
Green-light specs, HAND assembly, checkboxes	Charles	—
Slide visual design, Change Picture	Charles	—
Theory / manuscript integration later	VECTOR	When Act III done

SCOUT first deliverable when green-lit: Priority 1 PNG + JSON + 5-line README stub in `basic_data_plots/CCT_README.md` (what panel, what band, what to look for).

COMPASS standing job: Keep this file’s **Status** and Phase tables current after each SCOUT drop; file `SCOUT_report_to_COMPASS.md` summary when a phase closes.

7 — Slide / caption language (COMPASS draft — edit in HAND)

Slide A — P1 full vs +DFT (side-by-side)

Title: *Same talent, different pond — league vs draft-ecosystem*

Caption (binding-safe):
*At fixed PPM $z \in [1.5, 2.0]$: **full panel** — thicker teammate ponds $\rightarrow$ **higher** draft rates (Jackal > Squid). **+DFT subsample** (programs with ≥ 1 pick in window) — sign **flips**: mid-pond slightly > top-pond (~23% vs ~21%), overlapping uncertainty. Not proof of environment effects; consistent with congestion **ranking** story **within** draft ecosystems only.*

Slide B — Priority 3 Q4

Title: *On contenders, draft mass sits at the top of the roster*

Caption:
*Draft rate vs within-team roster percentile, faceted by team talent quartile. Q4 (highest T_j): top rank ~9.5% vs mid ~3.0%. Does not fix $\hat{A}$ — shows “big fish on **this** team” matters, especially on elite rosters.*

Speaker note for Alex:
Fix- A_i was the right move. Full league confounds program quality with pond thickness. The whisper may live in draft-ecosystem programs on PPM, plus within-team rank on contenders — not the old marginal hero curve.

8 — One paragraph for Alex (read aloud — post-sweep)

*We closed Act I and built the fix- A_i microscope you asked for. On the **full** NCAA panel at matched talent, players in **thicker** teammate ponds get drafted **more**, not less — elite-program confound. When we restrict to **programs that actually produce draft picks**, the sign **flips** on PPM: mid-pond slightly beats top-pond, but only by about two points with overlapping uncertainty. OBPM goes strongly the other way — appendix only. Separately, on **contender rosters**, draft mass concentrates at the **top of the team** — you have to stand out locally even when the team is elite. That’s our honest NCAA picture so far: not the marginal hero dip, but a draft-ecosystem whisper plus within-team rank on contenders.*

9 — Do-not-do (save campaign time)

- 1. **More ESPN seasons alone** — unlikely to reveal CCT (you + Alex agree).
- 2. **Another marginal poolq_loo ventile run** hoping bin 16 dips.
- 3. **Promoting OBPM/BPM to canonical hero** — appendix magnifier only.
- 4. ****Confusing T_j with poolq_loo**** in slides or prose.
- 5. **Claiming Army tiny K/N** — wrong; Army screams via macro tail drop.
- 6. **Claiming CCT confirmed from +DFT PPM alone** — ~2 pp, overlapping CIs, noisy ventiles.
- 7. **MERGE score + select + environment** in one sentence — BINDING violation.

10 — Collaboration model (three of you + two agents)

What's been working

- Porch reading as **single narrative**; technical memos as supplements.
- SCOUT **review file** with spot-checks (not silent edits).
- Charles **checkboxes only** Charles touches.

Tighter loop for this campaign (recommended)

Artifact	Purpose	Who updates
This file	Operational source of truth — priorities, checklist, specs	COMPASS
Porch reading	Stable story for Ginger / re-entry	Frozen unless science changes
<code>SCOUT_report_to_COMPASS.md</code>	End-of-phase: what shipped, paths, blockers	SCOUT after each Priority
<code>YYYYMMDD_SCOUT_CCT_campaign_review.md</code>	One-time sign-off on this plan (like porch review)	SCOUT
Question files	Only when blocked: <code>YYYYMMDD_COMPASS_to_SCOUT_questions.md</code>	COMPASS

What to stop doing

- Parallel memos that re-merge the same plot menu (porch + campaign + joint memo is enough).
- SCOUT building without Charles **Phase 0 green-light** on $\hat{A}$ band and axis.
- Long chat threads as spec — **spec lives in §3–4 of this file**.

Charles's workflow

1. Send SCOUT this file.
2. Lock Priority 1 spec (one message: band + axis).
3. SCOUT builds → drops PNG/JSON → pings in report.
4. You assemble HAND → brief Alex → mark checklist.

Optional later: Mirror a slim `.plan.md` to `plans/` if you want PDF via `convert_single_md_to_pdf.sh` — not required for day-one execution.

11 — Open decisions (Charles picks — defaults in parentheses)

Decision	Locked (2026-08-21)	Alternatives (later)
$\hat{A}$ band for Priority 1	$z \in [1.5, 2.0]$ ✓	Top decile; [1.0, 1.5] if bins thin
Primary pond axis	<code>poolq_loo</code> ventiles ✓	$\hat{T}_j$ ventiles (secondary panel)
Ventile count	16 (match hero)	10 wider bins at top
First HAND target	New CCT section	Append to next PD HAND

12 — Status log

Date	Event
2026-08-20	Act I BDP closed; porch reading published
2026-08-21	SCOUT porch review complete; Army/K/N clarifications merged
2026-08-21	Charles briefed Alex on fix $\hat{A}_i$ — Alex excited
2026-08-21	This campaign plan published — awaiting Charles green-light
2026-08-21	SCOUT campaign review complete — four engineering patches in §4
2026-08-21	Charles green-lit Priority 1 — <code>poolq_loo</code> , default $\hat{A}$ band; SCOUT building <code>pass_a_congestion_conditional.py</code>
2026-08-21	Act IIb sweep — +DFT PPM CCT YES (~2 pp); OBPM NO ; $\hat{T}_j$ rises; P3 Q4 top 9.5% vs mid 3.0%; see §3 scorecard
2026-08-21	COMPASS updated campaign plan + §13: Priority 2 heatmap next

13 — COMPASS sequencing (current)

Order	Plot	Status
1	P1 matched $\hat{A} \times \text{poolq_loo}$ (full + +DFT + OBPM + $\hat{T}_j$)	✓ Done — see §3 scorecard
2	P3 roster percentile $\times T_j$ quartile	✓ Done
3 (NEXT)	P2 heatmap — $\hat{A} \times \text{poolq_loo}$ (full + +DFT)	Awaiting Charles green-light
4	Wider $\hat{A}$ band / mg10 min10	After P2
5	P4 interval overlap + draft	Optional HAND enrich

Do not: Claim CCT confirmed from +DFT alone; promote OBPM to canonical; revert to marginal hero bin-16 dip hunting.
Repo hygiene: COMPASS checks `CCT_README.md`, `CCT_Campaign_Plan.md`, and `basic_data_plots/CCT_*` at start of each Charles chat.
SCOUT updates `CCT_README.md` on every drop.

Act II sweep done. HAND lead: PPM full vs +DFT + P3 Q4. Next build: Priority 2 heatmap.
— COMPASS · SCOUT sweep 2026-08-21 · P2 queued